🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 3 · STANDARD 6.AT.3

Ratio Reasoning: Convert Measurements Between Systems

Lesson 3-7 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

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My Notes Level 2 Standard

TODAY'S OBJECTIVES

Content Objective: I can use ratio reasoning to convert measurements between the customary and metric systems.

Language Objective: I can explain a conversion between systems using the words conversion factor, approximately, customary, and metric.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

Fill in each blank as we go. Use the Word Bank to help you.

WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Conversion factor | Customary system | Metric system | Approximately | Ratio table

Tap any word to see its meaning and a picture.

1. The ratio that trades one unit for another is a ________ ________.

2. Inches, pounds and gallons belong to the ________ system.

3. Centimeters, grams and liters belong to the ________ system, built on tens.

4. Close to but not exactly equal is ________ — written with ≈.

5. A table whose every column holds the same comparison is a ________ ________.

Write About the Math

Is 100 kilometers per hour faster or slower than 100 miles per hour, and how do you know?

¿100 kilómetros por hora es más rápido o más lento que 100 millas por hora, y cómo lo sabes?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Conversion factor** *Factor de conversión*
- ☐ **Customary system** *Sistema usual*
- ☐ **Metric system** *Sistema métrico*
- ☐ **Approximately** *Aproximadamente*
- ☐ **Ratio table** *Tabla de razones*

Model: *A mile is longer than a kilometer (1 km ≈ 0.6 mi), so 5 miles is the longer race even though the numbers match.* — Students reason about UNIT SIZE before touching arithmetic — same number, different units, different lengths.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The conversion factor is 1 km ≈ ____ mi. Scaling that up, 100 kilometers ≈ ____ miles, so the sign means about ____ miles per hour. That is ____ than 100 miles per hour, because in one hour you would travel ____ distance.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Conversion factor
2. **Notes 2:** Customary system
3. **Notes 3:** Metric system
4. **Notes 4:** Approximately
5. **Notes 5:** Ratio table
6. **Watch for this mistake:** *A common mistake converting between systems is comparing the bare numbers without converting at all — deciding that 5 kilometers and 5 miles are the same distance because both say 5. They are not: 5 kilometers ≈ 3 miles. A second mistake is writing = instead of ≈. Conversion factors between systems are rounded (1 mile is about 1.609344 kilometers), so the result is approximate no matter how carefully you multiply.*
7. **Try It 1:** *— Each pair is a conversion factor between the two systems, and each one is approximate.*
8. **Try It 2:** A. The conversion factors are rounded, so results are approximate *— One mile is about 1.609344 kilometers — a decimal that never ends neatly. Any usable factor is rounded, so the answer is approximate.*